

國立陽明交通大學

光電工程學系

碩士論文

Department of Photonics

National Yang Ming Chiao Tung University

Master's Thesis

結合編碼增強因子分解機與量子退火之矽光子反向設計

Inverse Design of Silicon Photonic Structures

Using Encoding-Enhanced Factorization Machines and Quantum Annealing

研 究 生： 蔡祐昕 (Wu, Yu Hsin)

指導教授： 吳致盛 (Wu, Jhih Sheng)

中華民國 一一五年八月

August 2026

結合編碼增強因子分解機與量子退火之矽光子反向設計

Inverse Design of Silicon Photonic Structures

Using Encoding-Enhanced Factorization Machines and Quantum Annealing

研 究 生： 蔡祐昕

Student： Yu Hsin Tsai

指導教授： 吳致盛 博士

Advisor： Dr. Jhih Sheng Wu

國立陽明交通大學

光電工程學系

碩士論文

A Thesis

Submitted to Department of Photonics

College of Electrical and Computer Engineering

National Yang Ming Chiao Tung University

in Partial Fulfillment of the Requirements

for the Degree of

Master of Science

in

Department of Photonics

August 2026

Taiwan, Republic of China

中華民國 一一五年八月

誌 謝

謝天謝地

蔡祐昕 於

國立陽明交通大學 光電工程學系

中華民國 一一五年八月



結合編碼增強因子分解機與量子退火之矽光子反向設計

學生：蔡祐昕

指導教授：吳致盛 博士

國立陽明交通大學
光電工程學系

摘 要

在過去文獻中，研究人員常使用基於梯度之伴隨方法 (Adjoint Method) 來設計光子結構。但此方法容易陷入局部最佳解，而難以探索全域最佳解。因此，本研究旨在結合機器學習與量子退火技術，並引入編碼方法以尋找全域最佳結構，為光子結構設計提供新的方法。本研究結合因子分解機 (Factorization Machines, FM) 與量子退火 (Quantum Annealing, QA) 技術，建構 FMQA 架構並應用於二維矽光子波導結構之反向設計 (Inverse Design)。

然而，受限於現有量子退火硬體之量子位元數量，本研究提出多元編碼方法，以提升設計參數之表示能力與設計空間解析度，同時也可以改善因結構離散化造成之量化誤差與階梯效應 (Staircase Effect)，確保數值模型與真實物理行為之高度一致性。在空間與特徵編碼方面，本研究具體採用了三種編碼策略：高斯上採樣 (Gaussian Upsampling)、二維傅立葉積分 (2D Fourier Integral) 以及變分自編碼器 (β -Variational Autoencoder, β -VAE)。

數值模擬的部分，本研究利用有限差分時域法 (Finite-Difference Time-Domain, FDTD) 針對橫電 (TE) 多模態在二維波導元件中之傳播行為進行分析，並將品質因子 (Figure of Merit, FoM) 設定為優化目標，以限制其低傳輸損耗，並提高傳輸效率與模態純度。

本研究設計並分析了兩種矽光子關鍵元件：90 度彎曲波導 (90-degree Bend Waveguide) 與緊湊型交叉波導 (Compact Waveguide Crossing)。在彎曲波導設計中，以降低

傳輸損耗並提升穿透率為主要目標；在緊湊型交叉波導中，則善用結構對稱性以減少量子位元需求，有效抑制模態轉換與串擾 (Crosstalk)，同時維持高傳輸效率。透過提出之 FMQA 架構與客製化目標函數，本研究成功推導出高效能且支援多模態傳輸的最佳化波導結構，為未來矽光子結構設計提供新的可能性與方向。

關鍵字：因子分解機、矽光子、波導、反向設計、量子退火、機器學習



Inverse Design of Silicon Photonic Structures Using Encoding-Enhanced Factorization Machines and Quantum Annealing

Student : Yu Hsin Tsai

Advisor: Dr. Jhih Sheng Wu

Department of Photonics
National Yang Ming Chiao Tung University

Abstract

In previous studies, researchers commonly used gradient-based adjoint methods (Adjoint Method) for the design of photonic structures. However, this method easily falls into local optima, making it difficult to explore globally optimal solutions. Therefore, this study aims to combine machine learning and quantum annealing techniques, and introduce encoding methods to search for globally optimal structures, providing a new approach for photonic structure design. This study integrates Factorization Machines (FM) and Quantum Annealing (QA) to construct an FMQA framework and applies it to the inverse design of two-dimensional silicon photonic waveguide structures.

However, due to the limited number of qubits in current quantum annealing hardware, this study proposes multi-encoding methods to enhance the representational capability of design parameters and the resolution of the design space. This approach also mitigates discretization-induced quantization errors and the staircase effect, ensuring high consistency between numerical models and real physical behavior. In terms of spatial and feature encoding, three encoding strategies are employed: Gaussian Upsampling, two-dimensional Fourier integral, and beta-Variational Autoencoder (β -VAE).

For numerical simulations, the finite-difference time-domain (FDTD) method is used

to analyze the propagation behavior of transverse electric (TE) multimodes in two-dimensional waveguide components. The figure of merit (FoM) is set as the optimization objective to constrain transmission loss and improve transmission efficiency and mode purity.

This study designs and analyzes two key silicon photonic components: a 90-degree Bend Waveguide and a Compact Waveguide Crossing. In the bend waveguide design, the primary objective is to reduce transmission loss and improve transmission efficiency. In the compact waveguide crossing, structural symmetry is exploited to reduce the number of required qubits, effectively suppress mode conversion and crosstalk while maintaining high transmission efficiency. Through the proposed FMQA framework and customized objective function, this study successfully derives high-performance optimized waveguide structures with support for multimode transmission, demonstrating new possibilities and directions for future silicon photonic integrated circuit design.

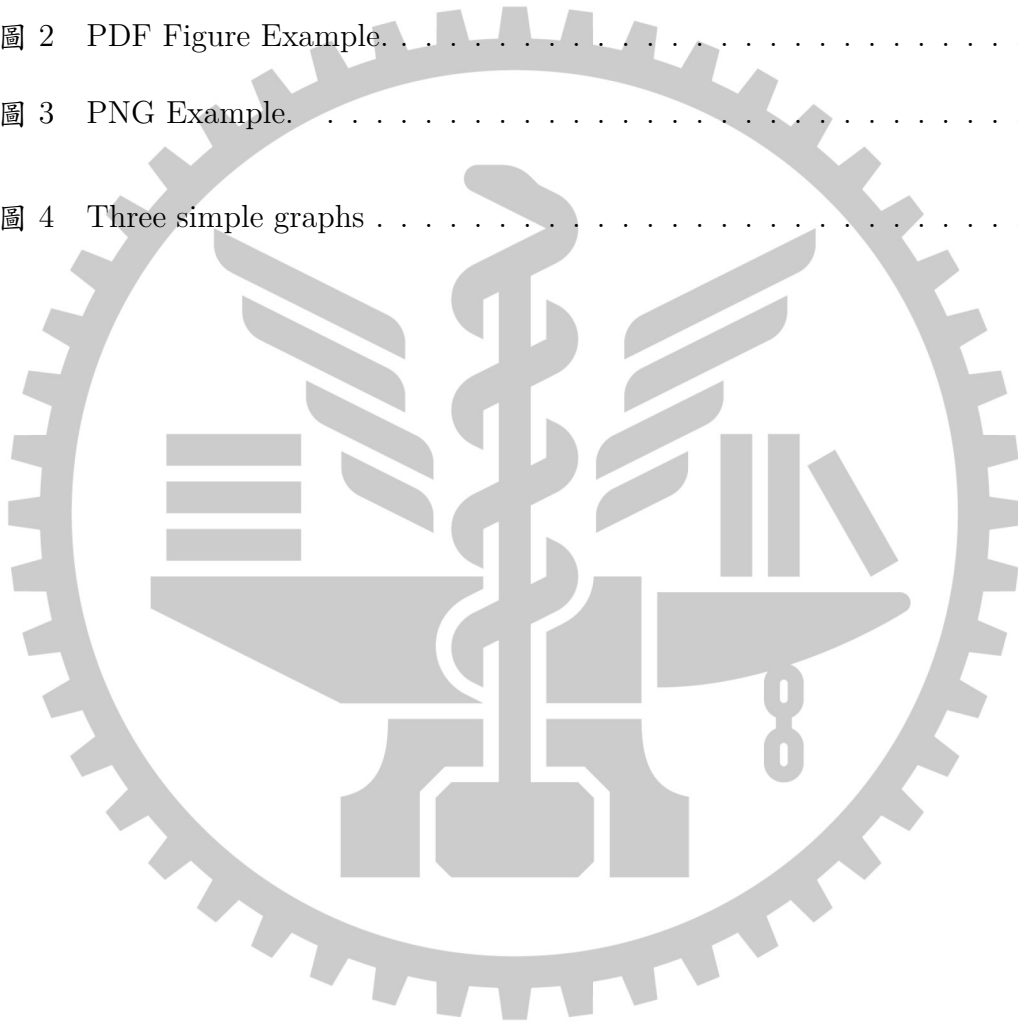
Keywords: Factorization Machine, Silicon Photonics, Waveguide, Inverse Design, Quantum Annealing, Machine Learning

目錄

中文摘要.....	i
英文摘要.....	iii
目錄.....	v
圖目錄.....	vi
表目錄.....	vii
第一章 Introduction.....	1
1.1 測試 0.....	1
1.1.1 測試 1.....	1
1.1.1.1 測試 2.....	1
第二章 Related Work.....	2
第三章 System Model.....	3
第四章 Data Fusion Algorithm.....	5
4.1 Data Preprocessing.....	5
4.1.1 2D LiDAR Data.....	6
第五章 Conclusions.....	7
第六章 Performance Evaluation.....	8
參考文獻.....	9
Appendix 附錄.....	11

圖目錄

圖 1	Excel2LaTeX.	4
圖 2	PDF Figure Example.	5
圖 3	PNG Example.	6
圖 4	Three simple graphs	8



表目錄

表 1	This is a table.	3
表 2	Comparison on model efficiency and model size.	3



第一章、Introduction

語法大幅度修改 (!?), 改成 xelatex 去編譯. 現在中文內容可以直接使用**粗體**跟*斜體*了.

1.1 測試 0

1.1.1 測試 1

還有研究一下 overleaf 支援的字體清單: <https://bit.ly/3MocQG3>

目前選擇 TW-Kai, 這個字體同時支援繁體與簡體中文, 有一些特殊字可以直接顯示, 像是之前有人問過的核苷酸.

1.1.1.1 測試 2

Video-based surveillance systems have been widely used in places such as plaza, office, factory, hotel, and conference hall for security purposes[1],[2].

The rest of this paper is organized as follows. Chapter 2 reviews some related work. Chapter 3 introduces our system architecture. Chapter 4 explains the details of our pairing algorithm. Performance evaluation results are in Chapter 5. Conclusions are in Chapter 6.

第二章、Related Work

通常第二段就是寫相關的參考文獻，只有 cite 到的文章才會出現編號並且出現在最後面。舉例來說，如果在 ref.bib 裡放了 10 篇論文，可是內文只有 cite 其中五篇，編譯出來的結果就只會顯示這五篇。Ref 有很多種風格寫法，本篇論文是採用 `bibliographystyle{IEEEtran}`，overleaf 上有其他 style 語法，可以參考：

https://www.overleaf.com/learn/latex/Bibtex_bibliography_styles

This is related work. The PID issue has been widely studied in the field of computer vision and IoT by using various devices. In the field of computer vision, camera is the most popular device. Face recognition technologies are surveyed in [3]. Reference [4] focuses on how to collect a very large training dataset and build a very deep CNN model for face recognition, but training process is extremely computationally expensive. A hybrid RFID and computer vision system for localization and tracking of RFID tags is proposed in [5]. Reference [6] presents a solution which combines RFID with object tracking through cameras. Reference [7] presents a fusion system consisting of an RFID reader and a camera crew on a mobile robot platform to track people. These works [5],[6],[7] fuse data from camera and RFID, but their accuracy highly depends on the density of RFID antennas. Thus, they are not suitable for longer range PID. Reference [8] proposes a fast multi-people tracking algorithm for service robots through RGB-D camera. In [9], people detection is realized by dense depth data, called Histogram of Oriented Depths (HOD).

第三章、System Model

如果想在 latex 裡面插入表格, 可以搜尋 latex table generator, 有很多線上網站可以參考. 我個人都是使用線上網站去產生大致的語法, 然後再根據個人喜好去做微調, wikibook 有很多資料可以參考, 網址在這邊: <https://en.wikibooks.org/wiki/LaTeX/Tables>

如果要引用表格, 記得在 table 裡加上 label 的語法, 然後就可以呼叫 Tab 1, 寫中文的就是表 1. 通常 Table 的 caption 是寫在表格的上面, 圖片則是放在下面.

表 1: This is a table.

A	1	4	7
B	2	5	8
C	3	6	9

後來在圖書館的“2022 研究攻略營論文寫作實戰技巧 (顏安孜老師)”看到另一種作法, 網址: <http://bit.ly/3yE06Hx>

裡面的講義有提到 Excel2LaTeX, 細節可以去看圖書館的連結, 裡面有放講義, 下方是顏安孜老師的講義截圖。(不過我個人偏好直接使用 latex 的表格語法, 這種轉檔的東西有時候會怪怪的)

表 2: Comparison on model efficiency and model size.

Model	Training & Test	Model Size
LTD-GCN	11ms & 4.5ms	2.50M & 2.50M
LTD-GCN-i	<u>13ms</u> & <u>4.6ms</u>	2.75M & 2.76M

如何製作表格

• 建議使用 Excel2LaTeX

```

498 Immediately following this sentence is the
499 point at which Table 1 is included in the
500 input file; compare the placement of the
501 table here with the table in the printed
502 output of this document.
503
504 \begin{table}[t]
505 \caption{Frequency of Special Characters}
506 \label{tab:freq}
507 \begin{tabular}{cc}
508 Non-English or Math & Frequency & Comments \\
509 \midrule
510 \pi & 1 in 1,000 & For Swedish names \\
511 \$ & 1 in 5 & Common in math \\
512 \Psi_1 & 4 in 5 & Used in business \\
513 \midrule
514 \Psi_1^2 & 1 in 40,000 & Unexplained usage \\
515 \bottomrule
516 \end{tabular}
517 \end{table}

```

Conference acronym 'XX, June 03–05, 2018, Woodstock, NY

Table 1: Frequency of Special Characters

Non-English or Math	Frequency	Comments
∅	1 in 1,000	For Swedish names
π	1 in 5	Common in math
\$	4 in 5	Used in business
Ψ ₁ ²	1 in 40,000	Unexplained usage

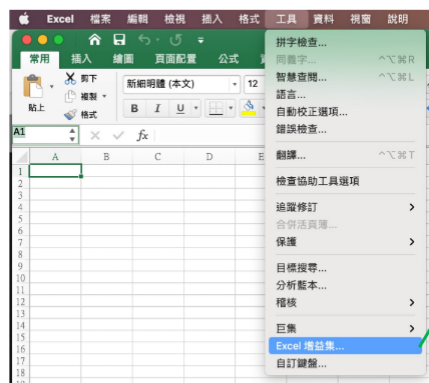
to be aligned properly in rows and columns, with the desired horizontal and vertical rules. Again, detailed instructions on **tabular** material are found in the *LaTeX User's Guide*.

Immediately following this sentence is the point at which Table 1 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page's

48

Excel2LaTeX



下載連結: <https://www.ctan.org/tex-archive/support/excel2latex/>

49

製作好表格之後，
框住表格範圍，
點擊增益集的按鈕，
即可產生表格語法

1. 框住表格範圍

2. 點擊增益集的按鈕

3. 點擊增益集的按鈕

4. 點擊增益集的按鈕

5. 點擊增益集的按鈕

50

圖 1: Excel2LaTeX.

第四章、Data Fusion Algorithm

這個是插入圖片的範例，圖片都放在 img 資料夾裡面。檔案格式有支援：JPG, PNG, PDF, EPS. 就使用自己習慣的繪圖工具，比較常見的應該就是 power point! power point 可以把繪圖區另存成 PDF, JPG, PNG, 還有 SVG. SVG 可以再轉成 PDF, 這樣圖片縮放還是會很清楚，可以把範例的兩張圖片都放大來看，應該可以看出差別。

最近常用的作法是，先使用 PPT 畫一張大圖，然後儲存成 PDF, 如果有白邊想去除的話，可以找線上網站裁切 (關鍵字: pdf crop free)

圖片出現的位置是由 latex 去決定，有時候會出現在奇怪的地方，這時候只能多爬文、嘗試各種參數，或者把整段圖片 code 放在前面試試看。

overleaf 上有插入圖片的介紹: https://www.overleaf.com/learn/latex/Inserting_Images

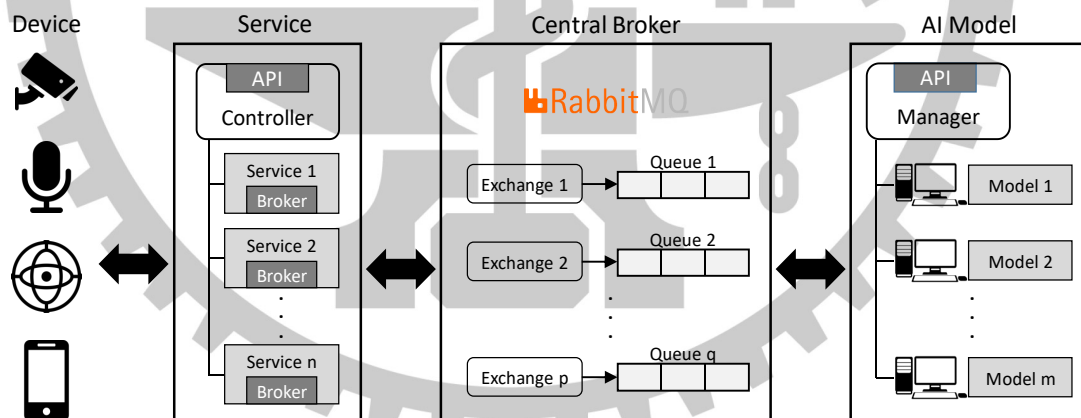


圖 2: PDF Figure Example.

4.1 Data Preprocessing

An example for section. Fig 2 is PDF. Fig 3 is PNG.

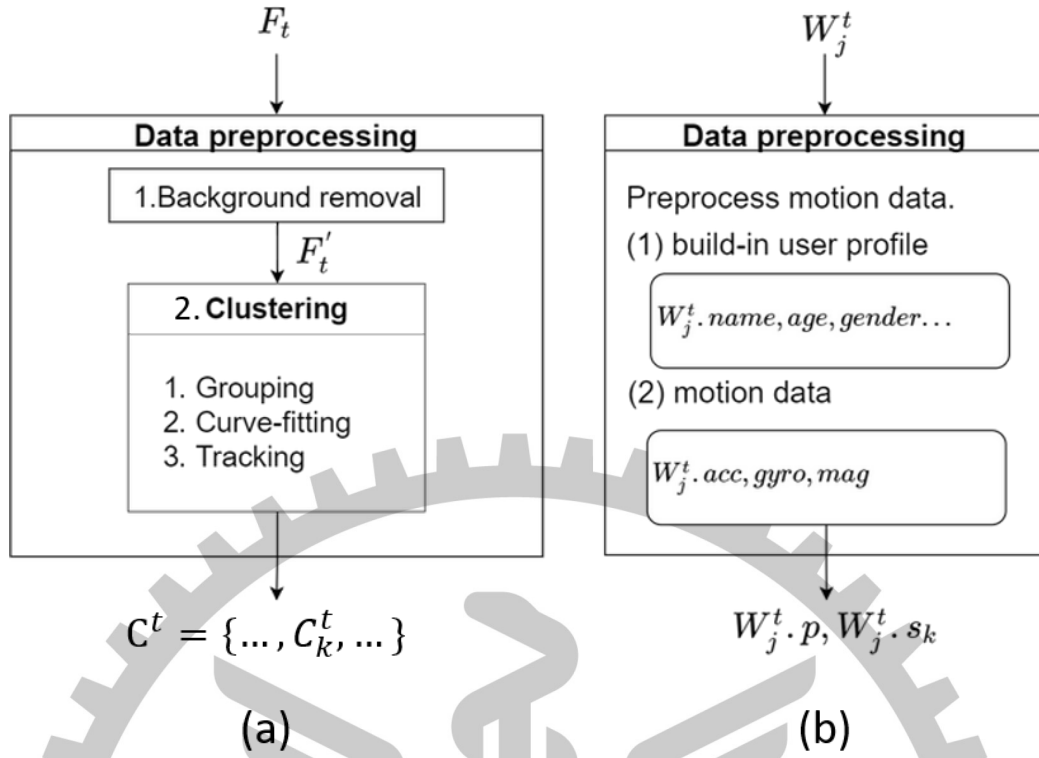


圖 3: PNG Example.

4.1.1 2D LiDAR Data

An example for subsection. 寫中文就是圖 2 跟圖 3.

第五章、Conclusions

Write your conclusion here.



第六章、Performance Evaluation

In this section, 整理效能評估.

下面是 subfigure 的範例 (其實我個人不常用這個語法, 我都直接在繪圖工具上把圖片整合在一起 XD)



圖 4: Three simple graphs

参考文献

- [1] R. T. Collins et al., “中文 a system for video surveillance and monitoring,” VSAM final report, pp. 1–68, 2000.
- [2] X. Wang, “Intelligent multi-camera video surveillance: A review,” *Pattern Recognition Letters*, vol. 34, no. 1, pp. 3–19, 2013.
- [3] W. Zhao, R. Chellappa, P. J. Phillips, and A. Rosenfeld, “Face recognition: A literature survey,” *ACM Computing Surveys*, vol. 35, no. 4, pp. 399–458, 2003.
- [4] O. M. Parkhi, A. Vedaldi, A. Zisserman et al., “Deep face recognition,” in *Proc. British Machine Vision Conference*, vol. 1, no. 3, 2015, p. 6.
- [5] M. Goller, C. Feichtenhofer, and A. Pinz, “Fusing rfid and computer vision for probabilistic tag localization,” in *Proc. IEEE International Conference on RFID*, 2014, pp. 89–96.
- [6] A. Isasi, S. Rodriguez, J. L. De Armentia, and A. Villodas, “Location, tracking and identification with rfid and vision data fusion,” in *Proc. European Workshop on Smart Objects: Systems, Technologies and Applications*, 2010, pp. 1–6.
- [7] T. Germa, F. Lerasle, N. Ouadah, and V. Cadenat, “Vision and rfid data fusion for tracking people in crowds by a mobile robot,” *Computer Vision and Image Understanding*, vol. 114, no. 6, pp. 641–651, 2010.

- [8] M. Munaro and E. Menegatti, “Fast rgb-d people tracking for service robots,” *Autonomous Robots*, vol. 37, no. 3, pp. 227–242, 2014.
- [9] L. Spinello and K. O. Arras, “People detection in rgb-d data,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2011, pp. 3838–3843.



Appendix 附錄

本附錄整理撰寫論文時常見的排版與語法注意事項，供大家撰寫與校稿時參考，請記得看 latex 的語法。(此段落主要由 ChatGPT 撰寫整理 XD)

文字與標點

- 英文引號請使用成對反引號與直引號：“text” → “text”。
- 破折號 (dash) 有三種：‘-’ (連字號，如 *pre-trained*)、‘—’ (範圍，如 10–20)、‘—’ (句中強調，如 the result—as expected—was correct)。
- 省略號請使用 `\ldots` 而非三個句點：A, B, C..., Z。
- 數值與單位間需留不換行空白：10 km, 30°C。
- 中英文混排時，中文與英文之間宜留一半形空白，例如：中文 English 中文。

數學與符號

- 數學變數自動為斜體，單位或文字常數須使用 `\mathrm{}` 或 `\text{}`，如 $v = 10 \text{ m/s}$ 。
- 三角函數、極值運算等應以數學命令表示： $\sin \theta$, $\max(x, y)$ 。

- 向量或矩陣可用粗體： $\mathbf{x} = [x_1, x_2]^\top$ 。
- 多行方程式請使用 `align` 環境，勿以多個 `equation` 拆寫。

圖表與引用

- 圖片標題（caption）置於圖下方
- 表格標題置於表上方。
- 參照圖表時請使用 `\ref{}` 或 `\autoref{}`，勿手動輸入編號。
例：As shown in Fig. 1.
- 表格建議使用 `booktabs` 套件：`\toprule`、`\midrule`、`\bottomrule`。

文獻與引用

- 文獻引用使用 `\cite{key}` 或 `\parencite{key}`。
- 多篇同時引用可寫為 `\cite{a,b,c} → [1-3]`。
- 特殊大小寫（如 GPU, LaTeX）須以大括號保留：`{GPU}`。